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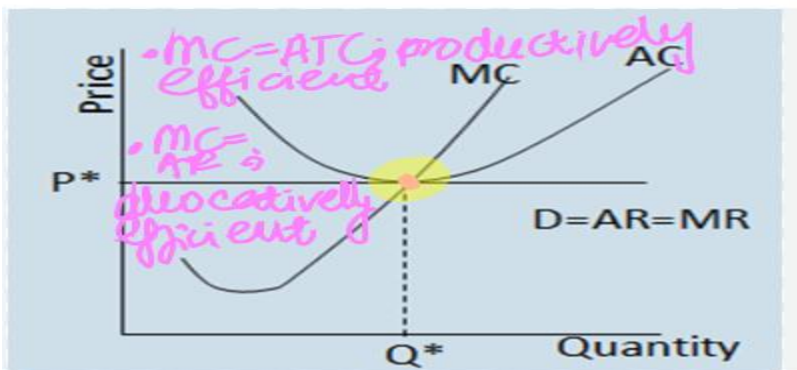
1-C:

- A- If the consumption of good has positive effects on third parties, this represents "external benefits"
- B – Private costs cannot exceed social costs. This is because $SC=PC+EC$
- C – If the production of a good has negative effects on third parties, this represents "external costs". This is why, key C, is correct.
- D – If social costs exceed social benefits, this represents "net social costs"

2-D:

When a perfectly competitive firm is maximizing its profits in the long-run, there is productive efficiency. This is because Average total cost is minimized since $MC=ATC$.

Similarly, since $MC=AR$ (price), this represents that allocative efficiency is achieved.



3-D:

- 3 The table shows the results of a cost-benefit analysis undertaken by a government when it was considering investing US\$200 million in building a new airport.

	US\$ million
private benefits	230
private costs	200
external costs	50

The government will build the airport if the net social benefit creates a return of at least 10% on its investment.

What will the minimum external benefit need to be in US\$ million to achieve this?

- A 10 B 20 C 30 D 40

Step 2
Investment $\Rightarrow 200$
Net social benefit $\Rightarrow 10\% \times 200 \Rightarrow 20$

Step #3

* $SB - SC \Rightarrow \text{Net social benefit}$
* $x - 250 = 20$
 $SB = 270$

Investment
Step #1
 $PC + EC = SC$
 $200 + 50 = 250$

Step #4
 $SB = PB + EB$
 $270 = 230 + x$
 $270 - 230 = x$
 $x = 40$

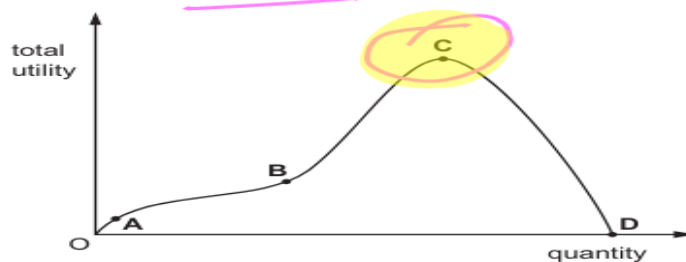
4-C:

This can be seen in the table below that when $MU=0$, TU is at its maximum.

Quantity	Marginal Utility (MU)	Total Utility (TU)
1	50	50
2	40	90 (50+40)
3	30	120 (90+30)
4	20	140 (120+20)
5	0	140 (no additional utility)

The diagram shows a total utility curve for a consumer.

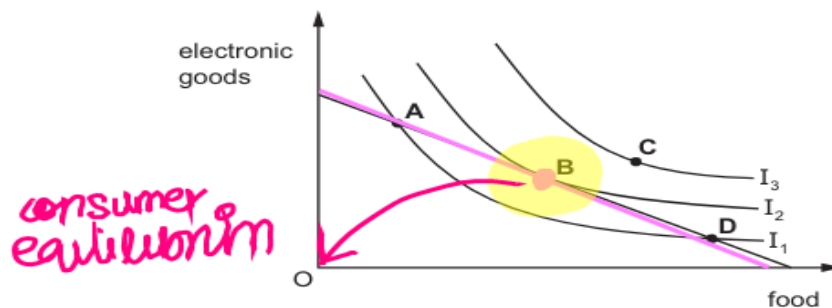
At which point does marginal utility equal zero?



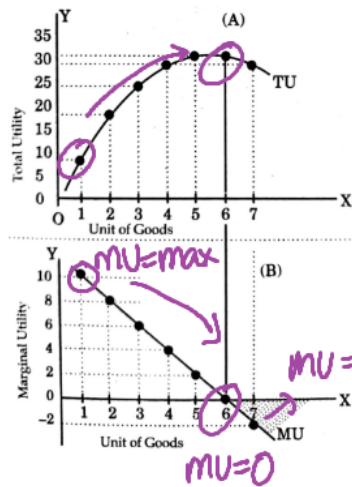
5B:

Consumer equilibrium represents the utility maximizing manner of a rational consumer. It is achieved when the indifference curve is tangential to the budget line. Here the equi-marginal principle that is when $MUX/PX = MUY/PY$ is established.

If income and prices remain the same, at which point on the diagram will the consumer maximise satisfaction?



6-A:



For the first unit consumed, marginal utility is at its maximum. At that point, total utility increases at an increasing rate.

As successive units are consumed, marginal utility diminishes which indicates that the law of diminishing marginal utility has set in. At that point, total utility increases at a decreasing rate.

When marginal utility is zero, total utility will be at its maximum.

When marginal utility is in negative, total utility falls.

This explains the shape of the total utility curve.

7-A:

A – Cheaper loans from bank as a firm grows large in size after a merger represent “financial economies of scale”, which comes under the category of internal economies of scale. This is why, key A, becomes correct.

B,C,D: Efficient local transport networks(key B), specialist training facilities provided by local college(key C) and the reputation of an area for a particular product(key D) represent external economies of scale.

8-B:

A) If the firm anticipates a rise in variable costs, it would likely reduce production or exit the market even in the short-run because there is a chance that it may fail to cover its variable costs. This is because the target of these firms is to cover variable costs in the short-run, since they form a significant proportion of total costs.

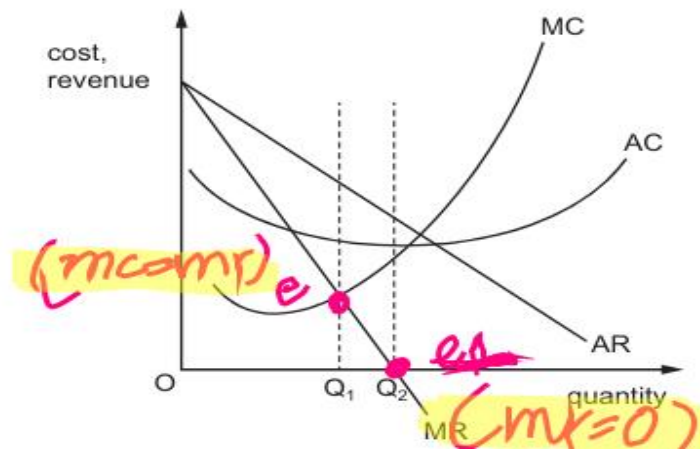
B) If the firm believes the price drop is temporary, it may continue producing to maintain market share so that it avoids losing customers. It expects to recoup losses when prices rise again.

C) Having large fixed costs represents a situation that fixed costs form the greatest proportion of the total costs of the firm. In that case, the firm may also give preference to the coverage of Average fixed costs.

D) If the firm has no control over the prices it can charge, this represents the case of perfect competition. In such a market structure, if subnormal profits are made, some firms leave the market. When supply decreases and prices rise again, normal profit equilibrium is achieved in the long-run.

9-C:

When output is increased from Q_1 to Q_2 , this represents that the objective changes from profit maximizing level of output ($MC=MR$) to sales revenue maximization ($MR=0$)



Because of this, the total profit will reduce, and total revenue will increase.

This explains why key C will be correct.

10-D:

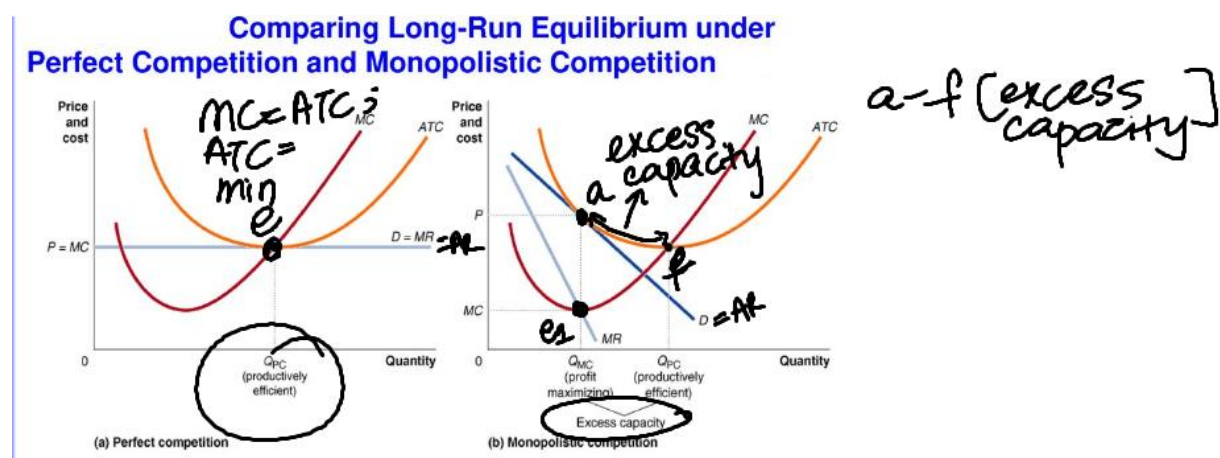
A: For economies of scale to be achieved, prices do not have to be increased.

B: Economies of scale does not occur because of management policy changes; rather, it is associated with a fall in long run average costs as a result of a firm growing in size.

C: Both, economies of scale and dis-economies of scale occur in the long-run.

D: This is correct because in case of economies of scale, there is a fall in long run average costs, as compared to diseconomies of scale in case of which, there is a rise in long-run average costs.

11-A:



A - Excess capacity in monopolistic competition refers to a situation where firms in the industry produce at a level that is below their optimal capacity, resulting in unused resources and inefficiency. This occurs because of producing less than the productively efficient level of output as can be seen in the diagram above. This explains why key A is correct.

B – Both, perfect competition and monopolistic competition achieve normal profits (zero economic profits) in the long-run. This means that none of them earns more than each other.

C – This is incorrect because there are a large number of firms in perfect competition (1000s of producers) as compared to monopolistic competition (20-25 small firms)

D – This is incorrect because there is a higher price elasticity of demand in case of perfect competition in the case of which the demand is perfectly elastic($ped = \infty$)

12-B:

B) When a single company dominates the market, and there are only a few players, it creates an oligopoly market structure. In this case, the largest company has the power to influence prices and market trends, and the relatively smaller companies may follow its lead, making it a price leader.

A) Monopolistic competition is a market structure characterized by many firms (20-25 small firms) producing differentiated products, which is not the case here.

C) Freedom of entry is not guaranteed, as the capital costs of entering the rail service market may still be high.

D) Ticket prices will not be guaranteed based on average costs. This is because, in this scenario where there is a dominant firm in the market, the smaller firms are most likely to follow the price leadership of that firm.

13-C:

C - Real wages refer to the actual value of wages in terms of the goods and services they can buy, taking into account the current price level. It's the purchasing power of money wages, adjusted for inflation or deflation. In other words, real wages represent the true value of the wages earned, considering the cost of living.

A) Marginal physical product of labor refers to the additional output produced by one more unit of labor.

B) The opportunity cost of labor refers to the value of the next best alternative that is given up when labor is used in a particular activity or job.

D) Wage net of tax refers to take-home pay, which is wages minus taxes, but doesn't reflect the purchasing power of those wages. Instead, they reflect disposable incomes of individuals.

14-D:

In the bar graph, there is a **fall** shown in the rate of average weekly earnings growth.

This can occur because of a fall in the demand of labor and a rise in the supply of labor.

- A- A decrease in the supply of labor, after having created shortages of labor, is most likely to result in an upward pressure on the rate of growth of earnings.
- B- A decrease in the interest rate will reduce costs of borrowing, as a result of which, investment activity on production by firms is likely to be stimulated. This will further lead to a rise in demand for labor to meet a rise in production, as a result of which, there will be an increase in the rate of growth of earnings.
- C- An increase in labor productivity will bring about a rise in marginal product of labor, because of which, the marginal revenue product will increase, and so, the demand curve shifts rightwards. This will be most likely to result in an upward pressure on the rate of growth of earnings.
- D- An increase in the size of the labor force will result in an increase in the supply of labor. This is most likely to result in a downward pressure on the rate of growth of earnings. As a result of which, key D, becomes correct.

15-A:

Indirect taxes are taxes on goods and services, examples include: sales tax.

They are regressive in nature, which means that they take up a greater proportion of incomes of the poor than the rich, because the average rate of tax decreases upon a rise in income.

Not only this but the marginal rate of tax is also less than the average rate of tax, which represents that for every additional \$1 earned, less is taken up in tax.

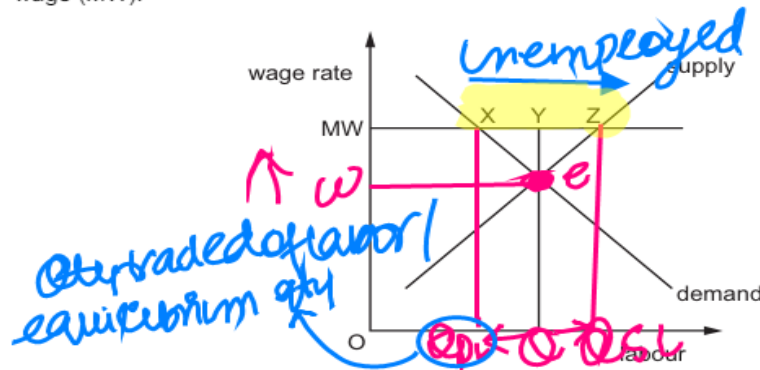
When indirect taxes are reduced, prices of goods and services fall, but since the consumption by low income groups on essential items extends by a relatively smaller percentage due to their demand being inelastic, this will bring about a fall in total expenditure of the poorer groups, which is why, they are benefitted.

Means-tested benefits are targeted towards those who need them the most. Examples include transfer payments such as: unemployment benefits and social security benefits. Increasing means-tested benefits would provide more financial support to low-income households, who are eligible for these benefits based on their income level.

Universal benefits (such as universal basic income, free healthcare, etc.) would provide a basic level of financial security to all citizens. Since they benefit the higher income groups as well, they can limit the redistributive effects. This is why, they should be reduced.

16-B:

- 16 The diagram shows the impact on the labour market of the introduction of a national minimum wage (MW).



Since because of a minimum wage legislation, when wages are raised to a level above the equilibrium, quantity demanded of labor or equilibrium quantity contracts.

Therefore, the distance from X-Y represents existing workers who are made redundant.

Similarly, because of this, quantity supplied of labor extends from Q to QSL. This represents the **unemployed new entrants, and is shown through the distance from Y to Z.**

17-B:

A) This journey already passes through the congestion charge area (area 1), so motorists traveling from X to Z are already paying the congestion charge. Extending the congestion charge area to include area 2 (point Y) doesn't affect this journey. Therefore, there is no new congestion charge applied, and no decrease in journeys is likely.

B) Previously, motorists traveling from X to Y didn't have to pay the congestion charge, as point Y was not in the congestion charge area. However, if the congestion charge area is extended to include area 2 (which includes point Y), then motorists traveling from X to Y will now have to pay the congestion charge. This new cost will likely discourage some motorists from making this journey, leading to a decrease in the number of journeys from X to Y. **This is why, this becomes correct.**

C) This journey already passes through the congestion charge area (area 1), so motorists traveling from Y to Z are already paying the congestion charge. Extending the congestion charge area to include area 2 (point Y) doesn't change the fact that this journey already passes through the congestion charge area. Therefore, there is no new congestion charge applied, and no decrease in journeys is likely.

D) This journey already passes through the congestion charge area (area 1), so motorists traveling from Z to Y are already paying the congestion charge. Extending the congestion charge area to include area 2 (point Y) doesn't affect this journey, as it doesn't change the fact that it already passes through the congestion charge area.

18-B:

A) While urbanization can lead to economic growth as it increases labor force in urban areas, it will not contribute to sustainable economic growth in a developed economy. Urbanization can put pressure on cities' infrastructure, services, and housing, leading to potential social and economic challenges.

B) Technical training is a key driver of sustainable economic growth. It enhances workers' skills, making them more productive and adaptable to technological changes. This leads to improved innovation and subsequently competitiveness driving economic growth and prosperity.

C) Population growth can lead to increased consumption and demand, but it also puts pressure on resources, infrastructure, and services.

D) While increased consumer credit can boost consumption and economic activity in the short term, it's not a sustainable driver of economic growth. High levels of consumer debt can lead to financial instability, reduced savings, and decreased investment in human capital and innovation in the long-run.

19-D:

The income method adds up all the income earned by factors of production, including wages, rents, interests, and profits, to calculate national income. Transfer payments, on the other hand, are not considered part of national income, as they do not represent new production or value added to the economy.

D - When calculating national income by the income method, transfer payments are not included. Transfer payments are payments made by the government to individuals or businesses without receiving any goods or services in return, such as social security benefits, pensions, and unemployment benefits. This is why, key D, becomes correct.

A) Income from abroad: Includes income earned by citizens and businesses from foreign investments and exports. These are considered when calculating gross national income.

B) Profits of private sector businesses: Includes profits earned by private sector businesses.

C) Rent from the ownership of land: Includes rental income earned by landowners.

20-A:

➔ Basic Income + Bonus Payment (W) = Gross Income

Basic Income: This is the regular income a person receives for their work or services.

Bonus Payment (W): This is an additional payment made to the individual, such as a performance bonus.

Gross Income: This is the total income a person receives, including their basic income and any **bonus payments**.

➔ **Gross Income - Direct Taxes (X) = Disposable Income**

Gross Income: This is the total income a person receives, as calculated above.

-Direct Taxes (X): These are taxes that are deducted directly from a person's income, such as income tax.

Disposable Income: This is the amount of income a person has available to spend or save after direct taxes have been deducted.

This explains why key A is correct.

21-C:

An increase in investment spending which represents an autonomous injection into the circular flow results in a multiplier times increase in real national income depending on the size of investment multiplier.

The formula of investment multiplier is $1/(1-mpc)$ or $1/mps$

Therefore, when households have a higher marginal propensity to consume, this would result in a rise in the size of the multiplier, because of which, there will be a larger increase in real national income. This explains why key C is correct.

22-B:

Let's first compare the HDI scores among countries.

Country B has the highest HDI score.

- A developed country is expected to have a high HDI score, indicating a high standard of living, educational attainment, and longevity.
- Based on the HDI scores, which range from 0 (least developed) to 1 (most developed), the rankings are:
 - B (0.897)
 - A (0.727)
 - C (0.527)
 - D (0.516)

Next we need to calculate gdp/capita of each country:

-> \$10.9 trillion = \$10,900 billion

Then, we can divide by the population in billions:

A) $\$10,900 \text{ billion} \div 1.37 \text{ billion} = \$7,956$

B) $\$2,400 \text{ billion} \div 0.0668 \text{ billion} = \$35,928$

C) $\$481 \text{ billion} \div 0.1822 \text{ billion} = \$2,640$

D) $\$19 \text{ billion} \div 0.0156 \text{ billion} = \$1,218$

It can be seen that country B has the highest gdp per capita

This explains why key B is correct.

23-A:

In this question, we are interested in an expansionary fiscal policy which is aimed at reducing cyclical unemployment.

A – An increase in government spending, as part of expansionary fiscal policy, after having raised Aggregate demand, will result in an increase in production and demand for labor, which represents that it will reduce cyclical unemployment.

B – An increase in government spending on education and training is more likely to be a supply side policy, which is aimed to improve occupational mobility of workers and reduce structural unemployment.

C – A reduction in tax on imports is likely to reduce domestic spending as now spending on imports will increase. This will dampen Aggregate demand, bringing about an increase in cyclical unemployment.

D – A reduction in indirect taxes is most likely to reduce costs of production and subsequently boost Aggregate supply.

24-C:

A- While countries X and Y experienced an upward trend in the rate of unemployment over the period, country Z experienced a downward trend in the rate of unemployment over the period.

B- We cannot deduce if country X had the highest number of people out of work because we are only given data in percentages.

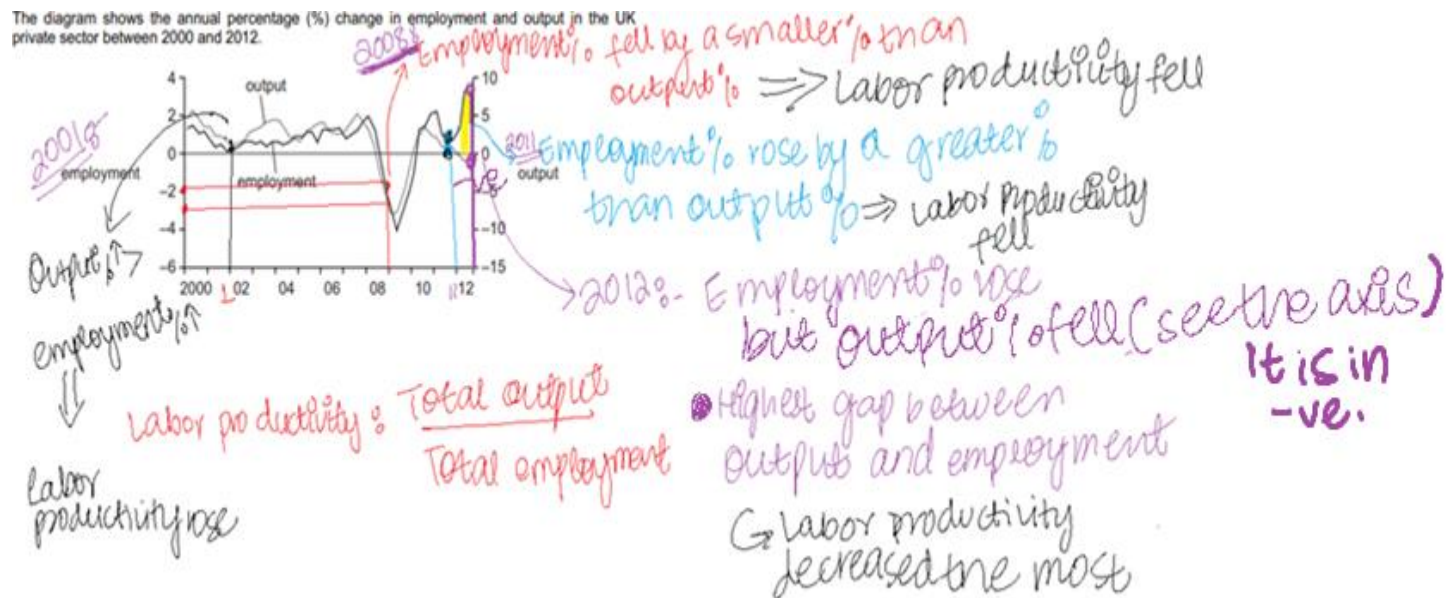
C- It can be seen that when there was recession in between 2008-2010, it was only country Y that saw a fall in the rate of unemployment when country X and Z experienced an increase in the rate of unemployment.

Therefore, it will be correct to say that the country which was least harmed by recession was country Y.

D - we cannot calculate the total number out of work in 2000 and 2010, because we are only given data in percentages.

25-D:

The diagram shows the annual percentage (%) change in employment and output in the UK private sector between 2000 and 2012.



To check the year in which labor productivity decreased the most, we will need to check labor productivity at the end of that year. The formula for labor productivity is output/employment.

26-A:

A) Since Greece has a smaller percentage of part time employees, this represents that it has a greater percentage of employees in full time employment as compared to the Netherlands. This is why, A becomes correct.

B) We can't conclude that from the information. While Greece has a longer average working week overall, we don't know the specific hours worked by part-time employees in each country.

C) Even though the overall average working week in Greece is longer, we can't say that the full time employees worked for longer hours than in the Netherlands. This is because, we don't know the specific hours worked by part-time employees in each country.

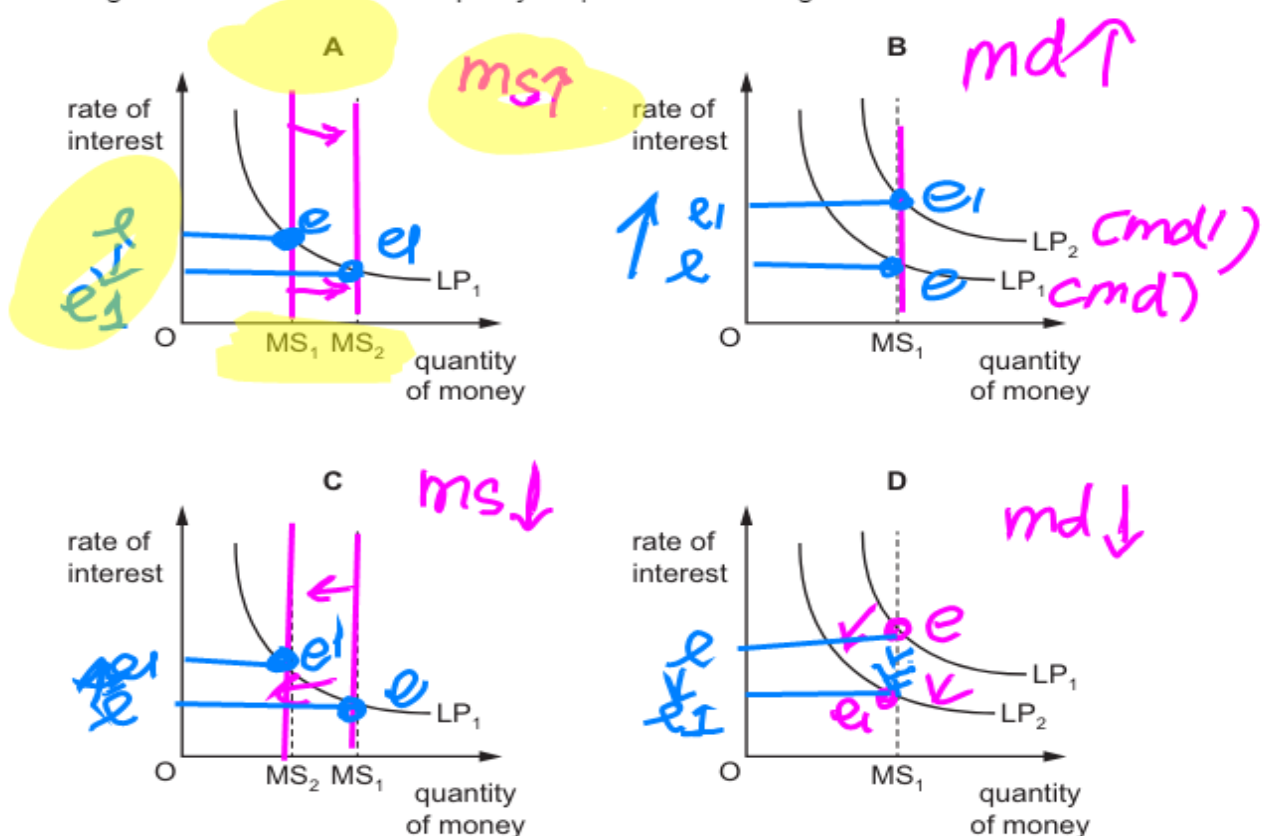
D) Based on data in percentages, we cannot deduce the number of part time employees in each country.

27-D:

- **Structural unemployment** occurs when jobs are lost due to changes in the structure of the economy, which means that there are changes in the pattern of demand and supply or there are technological advancements or shifts in global trade. In this case, the report suggests that automated robots could replace jobs in manufacturing and retail, indicating a structural change in the economy.
- **Demand-deficient unemployment** (A) occurs when there is a lack of demand for labor, typically during economic downturns or recessions. This is not the case in the paragraph, as the threat to jobs is due to automation, not a lack of demand.
- **Frictional unemployment** (B) refers to temporary unemployment due to the time it takes for workers to find new jobs or for employers to find suitable candidates. This type of unemployment is not related to the scenario described in the paragraph.
- **Seasonal unemployment** (C) occurs in industries with seasonal fluctuations, such as agriculture or tourism. This type of unemployment is not relevant to the paragraph's context.

28-A:

Which diagram shows the effect of a policy of quantitative easing on the rate of interest?



Quantitative easing represents a situation whereby the central bank buys back the securities previously sold to the non-bank private sector, on the behalf of the government.

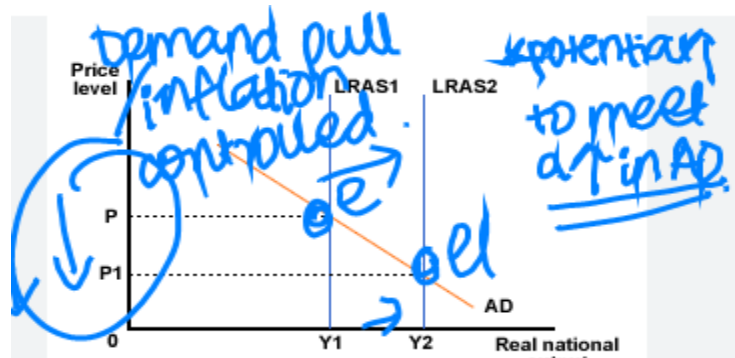
This results in an increase in money supply, and with an unchanged money demand, the interest rate or the price of money will fall.

This explains why key A will be correct.

29-D:

Changes in exchange rate are made as part of monetary policy, and both, deflationary monetary policy and fiscal policy are "short-run" measures aimed at dampening Aggregate demand and reducing inflationary pressures.

On the other hand, mainly interventionist supply side tools such as: education and training are long-run means of reducing demand pull inflation. This can be seen in the diagram below.



30-C:

C - When the economy is operating on the long-run aggregate supply curve, it means that it is at its maximum potential output, and any increase in aggregate demand will only lead to inflation, and there will be no growth in real GDP.

Consumption and investment expenditure being interest rate elastic (A) would actually make it easier for the government to achieve a faster rate of growth, as monetary policy (interest rates) could be used to stimulate the economy. This is because a fall in interest rates by even a smaller percentage would boost investment and consumption by a relatively higher percentage.

- The consumer price index being below its target (B) would suggest that there is room for expansionary policies to boost growth without worrying about inflation.
- A large negative output gap (D) would indicate that the economy has spare capacity, making it possible to achieve a faster rate of growth without hitting the maximum potential output.

